Multi-Year Assessment of Recreational Water Quality at New Jersey Beaches (2021–2025)

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Abstract

Recreational water quality monitoring helps protect public health by identifying bacterial contamination that may increase the risk of waterborne illness. *Enterococcus* bacteria are commonly used as an indicator of fecal contamination in marine recreational waters, and routine monitoring provides valuable information about the safety of public beaches. This study evaluated recreational water quality at New Jersey beaches using *Enterococcus* monitoring data collected during the recreational swimming seasons from 2021 through 2025.

The dataset was obtained from the New Jersey Department of Environmental Protection (NJDEP) and analyzed in R. After data cleaning, descriptive statistics were calculated, and annual, seasonal, and geographic trends were evaluated. Data visualizations were created to examine the distribution of *Enterococcus* concentrations, and logistic regression was used to assess factors associated with samples exceeding the U.S. Environmental Protection Agency (EPA) recreational water quality standard.

A total of 16,745 water samples were included in the analysis. Most *Enterococcus* measurements were well below the EPA threshold, with only 78 samples (0.47%) exceeding 104 CFU/100 mL. The percentage of exceedances varied somewhat by year and increased across the swimming season, with the highest monthly percentage occurring in September. Descriptive differences were also observed among beaches, although exceedances remained uncommon overall. These results indicate that measured *Enterococcus* concentrations were generally below the selected threshold across the monitored New Jersey beaches during 2021–2025, while still showing some variation across time and location.

1. Introduction

Recreational water quality is an important public health concern because contaminated water can expose swimmers and other beach visitors to microorganisms associated with gastrointestinal and other illnesses. Routine monitoring provides a way to identify periods when bacterial concentrations are elevated and when swimming conditions may require additional attention.

Enterococcus is commonly used as an indicator of fecal contamination in marine recreational waters.[1] Although the presence of *Enterococcus* does not necessarily mean that a person will become ill, elevated concentrations can indicate the possible presence of other microorganisms associated with fecal contamination. The EPA therefore uses *Enterococcus* concentrations as part of its recreational water quality criteria.[2] Beaches that exceed the relevant criteria may be subject to swimming advisories or temporary closures.

New Jersey's Atlantic coastline contains hundreds of recreational beaches that are monitored during the swimming season by the NJDEP.[3] The resulting monitoring data provide measurements of *Enterococcus* concentrations across a large number of locations and sampling dates.

The purpose of this study was to examine *Enterococcus* measurements collected at New Jersey recreational beaches from 2021 through 2025. The study examined differences among years, months, and beach locations and used logistic regression to evaluate factors associated with samples exceeding the EPA threshold.

2. Methods

2.1 Data Source

Water quality monitoring data were obtained from the NJDEP through its publicly available recreational water quality monitoring program.[3] The dataset included *Enterococcus* measurements collected from recreational beaches throughout New Jersey during the 2021–2025 swimming seasons. Each record contained information on sampling date, beach location, *Enterococcus* concentration (CFU/100 mL), and additional monitoring information reported by the NJDEP. The original dataset contained 16,816 observations collected from 295 recreational beaches.

2.2 Data Preparation

All data cleaning and analysis were performed in R (version 4.5.3) using RStudio. The dataset was imported as a comma-separated values file and examined for missing or incomplete observations. Entries with missing *Enterococcus* measurements were removed, resulting in a final dataset of 16,745 observations.

Sampling dates were converted to a standard date format, and additional variables representing the sampling year and month were created to support temporal analyses. For the seasonal analysis, months were grouped into Early Summer (May–June), Peak Summer (July–August), and Late Summer (September). A binary outcome variable was also created to classify samples as safe or unsafe based on the EPA recreational water quality criterion of 104 colony-forming units (CFU) of *Enterococcus* per 100 mL. Samples with *Enterococcus* concentrations greater than 104 CFU/100 mL were classified as unsafe for statistical analyses.

2.3 Statistical Analysis

Descriptive statistics were calculated for *Enterococcus* concentrations, including the mean, median, standard deviation, and percentage of samples exceeding the EPA threshold. The data were then summarized by year, month, and beach to examine differences in water quality across time and location.

The distribution of *Enterococcus* concentrations and patterns in exceedances were visualized using ggplot2. Figures were used to examine the overall concentration distribution, annual percentage of unsafe samples, monthly percentage of unsafe samples, and beaches with the highest exceedance percentages.

Logistic regression models were used to examine associations between threshold exceedance and sampling year, seasonal period, and beach location. Year was modeled as a numeric predictor to assess the overall linear temporal trend in exceedance odds, while seasonal period was modeled categorically using Early Summer (May–June) as the reference category. Odds ratios and 95% confidence intervals were calculated for interpretable model estimates. Because unsafe observations were rare and distributed across a large number of beach categories, beach-level estimates were evaluated for stability before interpretation. All statistical analyses were conducted in R using the dplyr and ggplot2 packages.

3. Results

3.1 Descriptive Statistics

A total of 16,745 water samples collected from 295 recreational beaches were included in the final analysis following data cleaning. The mean *Enterococcus* concentration was 8.08 CFU/100 mL, while the median concentration was 5.00 CFU/100 mL, indicating that most samples contained relatively low bacterial concentrations. The standard deviation was 16.27 CFU/100 mL, reflecting variability in *Enterococcus* measurements across samples.

Only 78 samples (0.47%) exceeded the EPA recreational water quality threshold of 104 CFU/100 mL, indicating that the vast majority of water samples met recommended recreational water quality standards during the study period.

3.2 Distribution of *Enterococcus* Concentrations

The distribution of *Enterococcus* concentrations is shown in Figure 1. Most water samples contained relatively low bacterial concentrations, with the majority of observations occurring below 20 CFU/100 mL. As *Enterococcus* concentrations increased, the frequency of observations declined sharply, resulting in a strongly right-skewed distribution.

The EPA recreational water quality threshold (104 CFU/100 mL) is indicated by the dashed vertical line in Figure 1. Only 78 samples (0.47%) exceeded the selected threshold of 104 CFU/100 mL, indicating that exceedances were uncommon in the analyzed dataset.

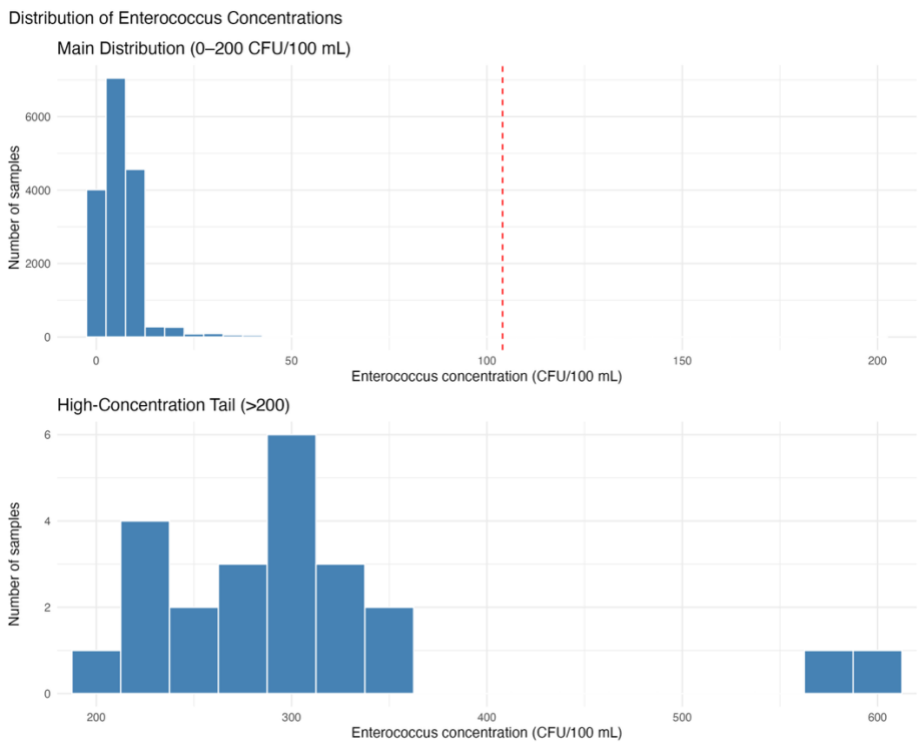


Figure 1. Distribution of *Enterococcus* concentrations measured at New Jersey recreational beaches during the 2021–2025 swimming seasons. The upper panel displays concentrations from 0 to 200 CFU/100 mL, with the dashed red line indicating the selected threshold of 104 CFU/100 mL. The lower panel displays concentrations exceeding 200 CFU/100 mL using a separate horizontal scale. Both panels include observations from the cleaned analytical dataset.

3.3 Annual Trends

The percentage of samples exceeding the EPA standard varied across the five years but remained below 1% in every year (Figure 2). The exceedance percentage increased from 2021 through 2023, when it reached approximately 0.73%, and then decreased to approximately 0.30% in 2024. In 2025, the percentage increased slightly to approximately 0.33%. Thus, 2023 had the highest proportion of samples exceeding the threshold, although exceedances were uncommon in every year. A logistic regression model treating year as a numeric predictor estimated a decrease of approximately 8.5% in the odds of threshold exceedance per year (OR = 0.915, 95% CI: 0.779–

1.071, p = 0.269). However, this trend was not statistically significant, as the confidence interval included 1.

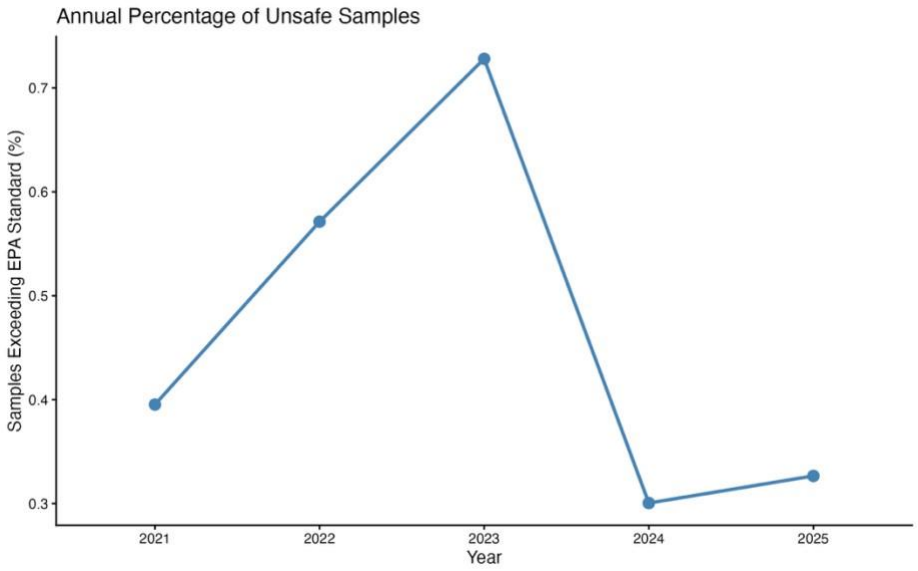


Figure 2. Annual percentage of water samples exceeding the EPA recreational water quality threshold during the 2021–2025 recreational swimming seasons. Although modest year-to-year variation was observed, the percentage of exceedances remained below 1% in every year.

3.4 Seasonal Trends

The percentage of samples exceeding the EPA standard increased from May through September (Figure 3). The lowest percentage occurred in May, followed by progressively higher percentages in June, July, August, and September. This pattern was also supported by logistic regression, which showed significantly higher odds of an exceedance during Peak Summer and Late Summer compared with Early Summer. The monthly exceedance percentage remained below 1% throughout the monitoring season, despite increasing from May through September.

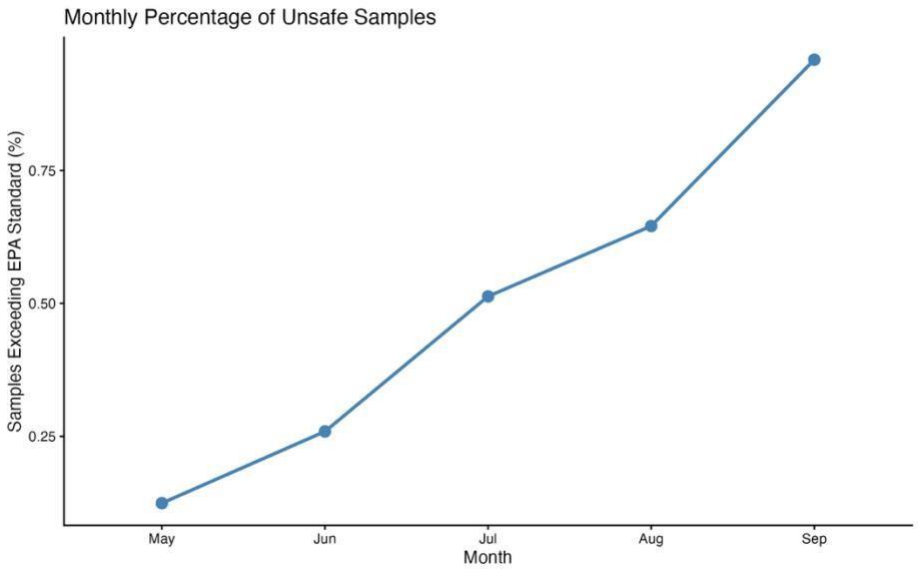


Figure 3. Monthly percentage of water samples exceeding the EPA recreational water quality threshold during the 2021–2025 recreational swimming seasons. The percentage of exceedances increased progressively from May through September, although fewer than 1% of samples exceeded the threshold in every month.

3.5 Beach Comparisons

The percentage of samples exceeding the selected threshold varied among the monitored New Jersey beach locations (Figure 4). The four locations with the highest observed exceedance percentages had rates of approximately 2%, while the remaining locations shown in the top ten had rates of approximately 1%. Some beach names, including South Carolina and North Carolina, resemble the names of other U.S. states but refer to locations in the New Jersey monitoring dataset. These differences should be interpreted cautiously because threshold exceedances were uncommon. Overall, exceedances occurred at 49 beach locations, and most locations had only one exceedance during the study period. Because the number of exceedances was small relative to the number of locations, these descriptive percentages should not be interpreted as definitive evidence of differences in beach-level risk.

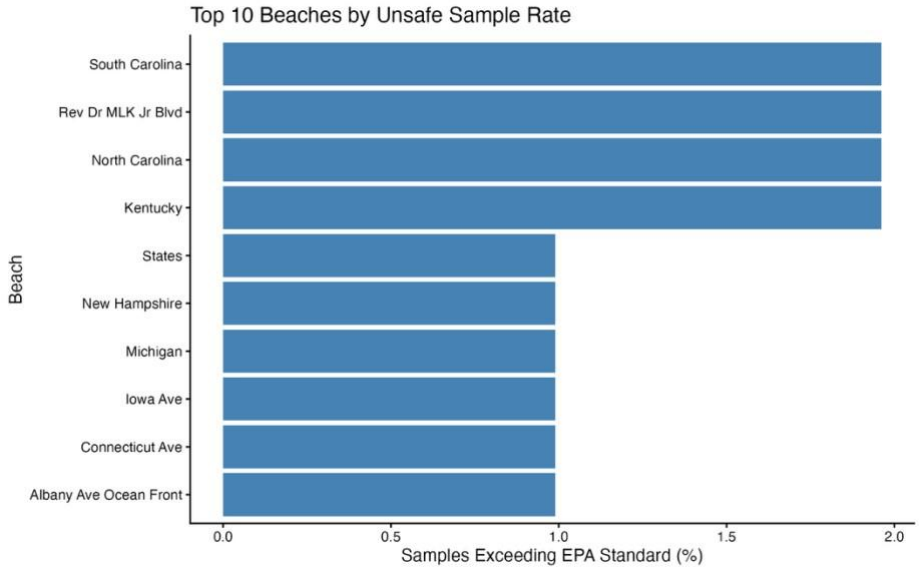


Figure 4. Ten New Jersey recreational beach locations with the highest observed percentages of samples exceeding the selected *Enterococcus* threshold during the 2021–2025 swimming seasons. The displayed percentages are descriptive and should be interpreted cautiously because threshold exceedances were uncommon across the dataset.

3.6 Logistic Regression

Logistic regression was used to evaluate whether the odds of an *Enterococcus* concentration exceeding the selected threshold varied over time and across seasonal periods. In the numeric-year model, the estimated odds of threshold exceedance decreased by approximately 8.5% per year (OR = 0.915, 95% CI: 0.779–1.071, p = 0.269). This trend was not statistically significant, indicating insufficient evidence of a linear change in exceedance odds across the study period.

Sampling season was significantly associated with threshold exceedance. Using Early Summer (May–June) as the reference period, samples collected during Peak Summer (July–August) had 2.77 times the odds of exceeding the threshold (OR = 2.768, 95% CI: 1.573–5.204, $p < 0.001$). Samples collected during Late Summer (September) had 4.59 times the odds of exceeding the threshold compared with Early Summer (OR = 4.592, 95% CI: 2.166–9.736, $p < 0.001$). Both confidence intervals were entirely above 1, indicating statistically significant associations between

seasonal period and threshold exceedance. These findings show that exceedances were more likely later in the monitoring season, although the analysis does not establish the cause of this pattern.

Exceedances occurred at 49 beach locations. South Bath had the greatest number of exceedances (n = 3), while 17 locations had two exceedances each and 31 locations had one each. Because only 78 of the 16,745 observations exceeded the threshold and these observations were distributed across many locations, beach-level logistic regression estimates were unstable and were not interpreted as reliable evidence of differences among individual beaches.

Overall, the regression analysis identified a statistically significant association between seasonal period and threshold exceedance, but it did not identify a statistically significant linear temporal trend across 2021–2025. These findings suggest that seasonal variation was more evident than a consistent year-to-year trend in the observed exceedances.

4. Discussion

The results show that *Enterococcus* concentrations were generally low across the New Jersey beaches included in this dataset. Only 78 of the 16,745 samples exceeded the EPA threshold, corresponding to 0.47% of all observations. Most measurements were therefore far below the threshold, although the distribution was strongly right-skewed because a relatively small number of samples had substantially higher concentrations.

There was some variation across years. The percentage of samples exceeding the EPA standard increased from 2021 to 2023 and then decreased in 2024, with a small increase again in 2025. The annual differences were relatively small compared with the overall number of samples, but the pattern shows that exceedances were not distributed evenly across the five-year period.

The monthly results showed a clear descriptive pattern: the percentage of samples exceeding the selected threshold increased from May through September, with September having the highest percentage. The seasonal logistic regression also found higher odds of threshold exceedance during Peak Summer and Late Summer than during Early Summer. However, the observed pattern does not establish why exceedances were more common later in the swimming season. Environmental conditions, including rainfall, water temperature, and tides, could potentially contribute to this variation, but these factors were not evaluated in the present analysis. The percentage of samples

exceeding the threshold remained below 1% in every month, so the seasonal pattern represents changes in the frequency of relatively uncommon exceedances rather than evidence of a general decline in water quality.

The beach-level results also showed descriptive differences among locations. The beaches with the highest exceedance percentages had rates of approximately 2%, compared with approximately 1% for the other beaches shown in the top ten. However, these percentages were based on relatively few unsafe observations. Across the full dataset, the 78 unsafe samples were distributed across 49 beach locations, and most locations had only one unsafe sample. Because of this sparse distribution, individual beach-level logistic regression estimates were unstable and were not interpreted as evidence of differences in beach-level risk.

The concentration distribution provides additional context for these results. Most samples were concentrated at relatively low *Enterococcus* values, while higher measurements occurred much less frequently. This type of distribution means that the mean alone does not fully describe the dataset; the median of 5.00 CFU/100 mL provides a useful indication of the concentration observed in a typical sample.

Several limitations affect how these findings should be interpreted. First, the dataset was collected through routine monitoring, and the analysis did not account for environmental conditions such as rainfall, temperature, or tides at the time of sampling. These factors could potentially contribute to variation in *Enterococcus* concentrations. Second, differences in sampling frequency among beaches or months, if present, could influence the observed exceedance percentages because the analysis summarized individual samples rather than equally weighting each beach or sampling period. Consequently, the results may reflect both differences in measured water quality and the distribution of sampling observations. In addition, *Enterococcus* was the only water-quality measure examined in this analysis. Finally, the data covered the routine recreational swimming seasons from May through September and therefore do not describe conditions outside that monitoring period.

Despite these limitations, the dataset provides a large number of observations across multiple years and locations. The combination of descriptive analysis, visualization, and logistic regression made it possible to examine both the overall distribution of *Enterococcus* concentrations and differences in exceedance rates across time and location. Future analyses could incorporate

environmental measurements, including rainfall, water temperature, and tidal conditions, alongside sampling frequency and location-specific characteristics to investigate potential explanations for the observed seasonal pattern and improve comparisons among beaches.

5. Conclusion

This analysis of New Jersey coastal beach water-quality data from 2021 through 2025 found that *Enterococcus* exceedances of the 104 CFU/100 mL threshold were uncommon, occurring in 0.47% of the 16,745 samples analyzed. The frequency of exceedances varied somewhat among years, but logistic regression did not identify a significant association between sampling year and the likelihood of an exceedance. In contrast, a significant seasonal pattern was observed.

Compared with Early Summer, the odds of an exceedance were approximately 2.8 times higher during Peak Summer and 4.6 times higher during Late Summer.

Descriptive differences were also observed among beach locations, but unsafe samples were distributed across 49 beaches, with most locations having only one unsafe sample. Because exceedances were relatively rare and spread across a large number of beach categories, individual beach-level regression estimates were unstable and were not interpreted as evidence of differences in beach-level risk. Overall, the results suggest that seasonal timing was more strongly associated with *Enterococcus* exceedances than year-to-year variation, while the low overall exceedance rate indicates that concentrations above the selected threshold were uncommon across the dataset. Future analyses incorporating environmental factors such as rainfall, temperature, tides, and other site conditions could help explain the seasonal pattern observed in this study.

References

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